



Decoding Nike's Customer Sentiment on Reddit

A Machine Learning Approach to understanding the Customer

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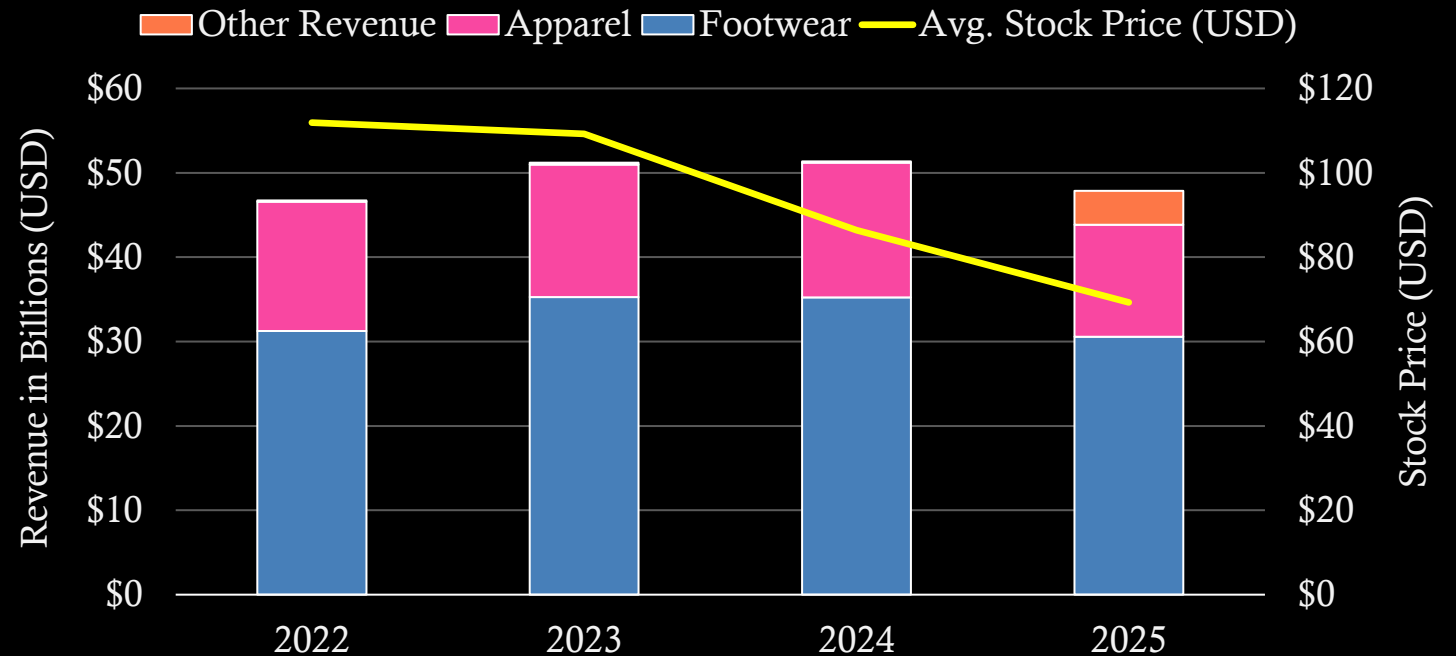
Business Problem

The Challenge of Trust

Business Problem

Nike is a globally recognized brand facing declining stock prices since late 2021. As the data team at Nike, we understand there are factors out of our control (trade laws, supply chain disruptions, and increased competition). However, with a shift in focus, we can lead product innovation, customer satisfaction, sales, and bring back the trust of our investors. But what do we focus on? As Nike's analysts, we turned to machine learning techniques to analyze customer online sentiment (Reddit). The goal is to better understand consumer preferences, identify areas for improvement, and inform product development and marketing strategies that boost satisfaction, sales, and shareholder confidence.

Nike Stock Price vs. Revenue By Sector



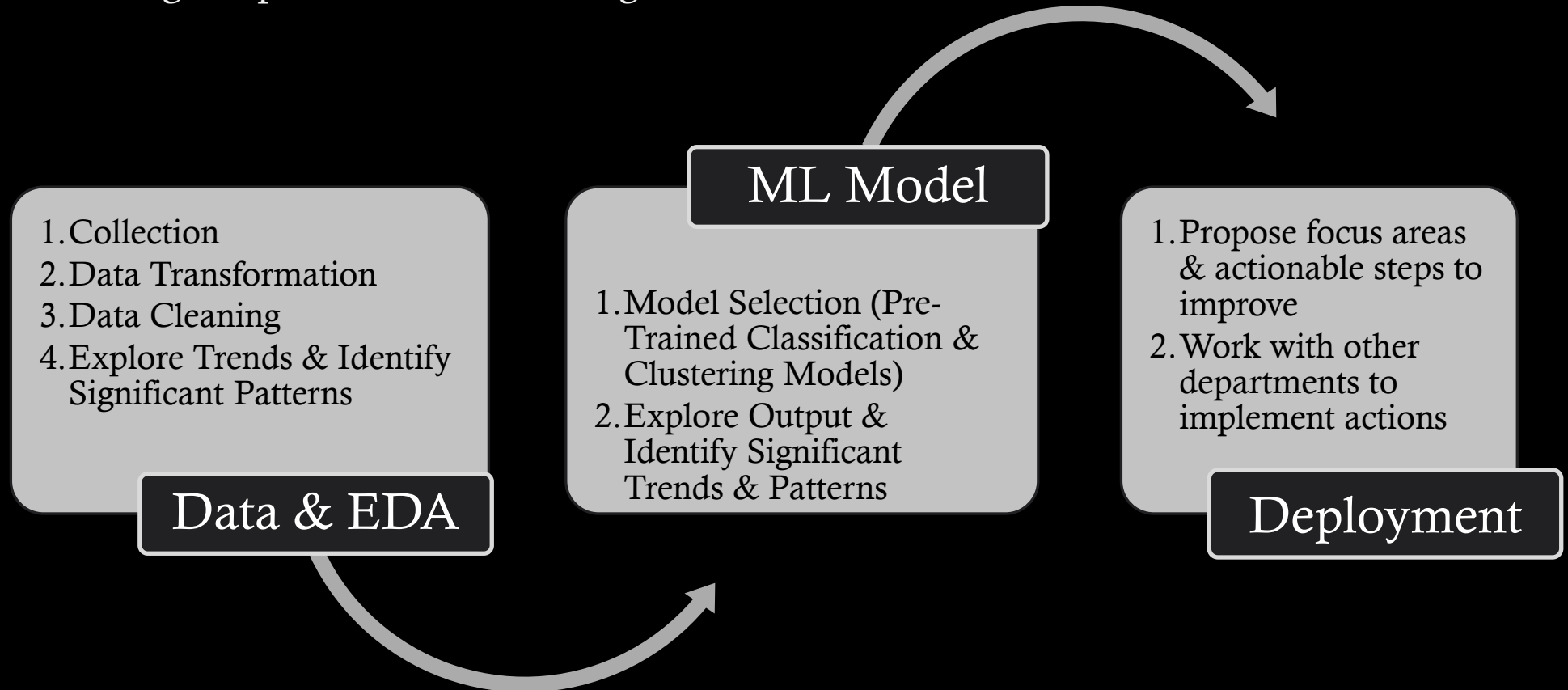
¹(NIKE, Inc. (NKE) Revenue by Segment)

Approach

A Machine Learning Approach to understanding the Customer

Approach

Our approach follows the CRISP-DM method where we evaluate the problem, understand the data, prep the data, run a model(s), evaluate, and deploy. Below are the detailed steps following our problem understanding.

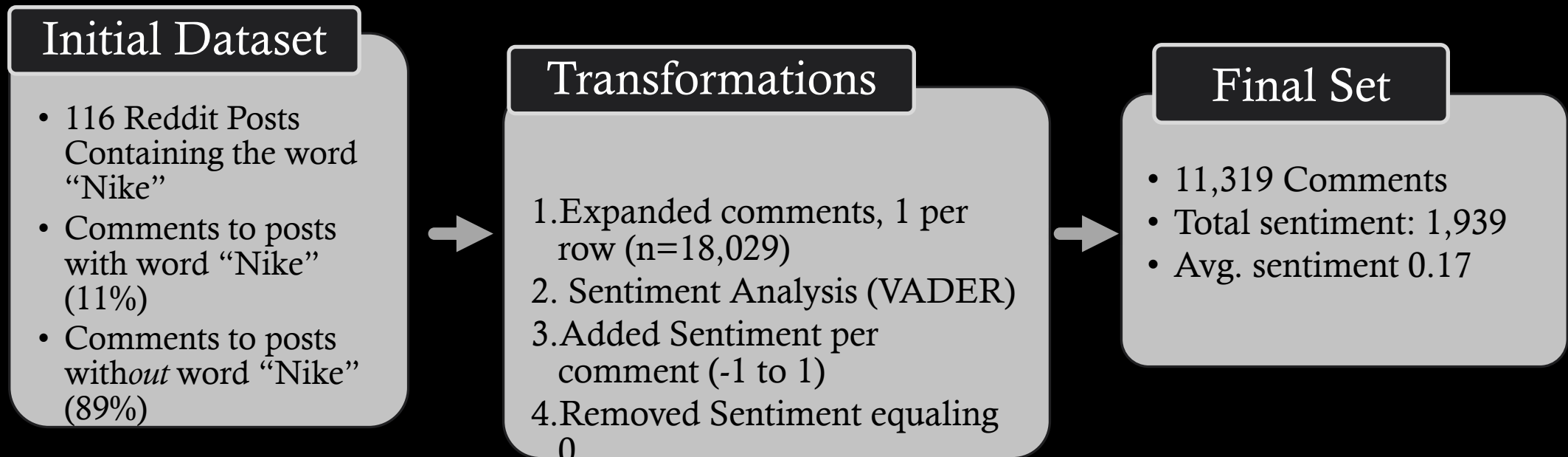


Data

18,000 Comments

Data

The data use to find what costumers are looking for was sourced from Reddit threads (titles & comments) mentioning 'Nike' from 2022 to March 2025. 116 posts collected, with an average of 152 comments per post. By using Reddit as opposed to other social media websites, we were able to avoid bias from paid marketing & influence.



Model Selection & Evaluation

A Pre-Trained Model Approach

Model Selection

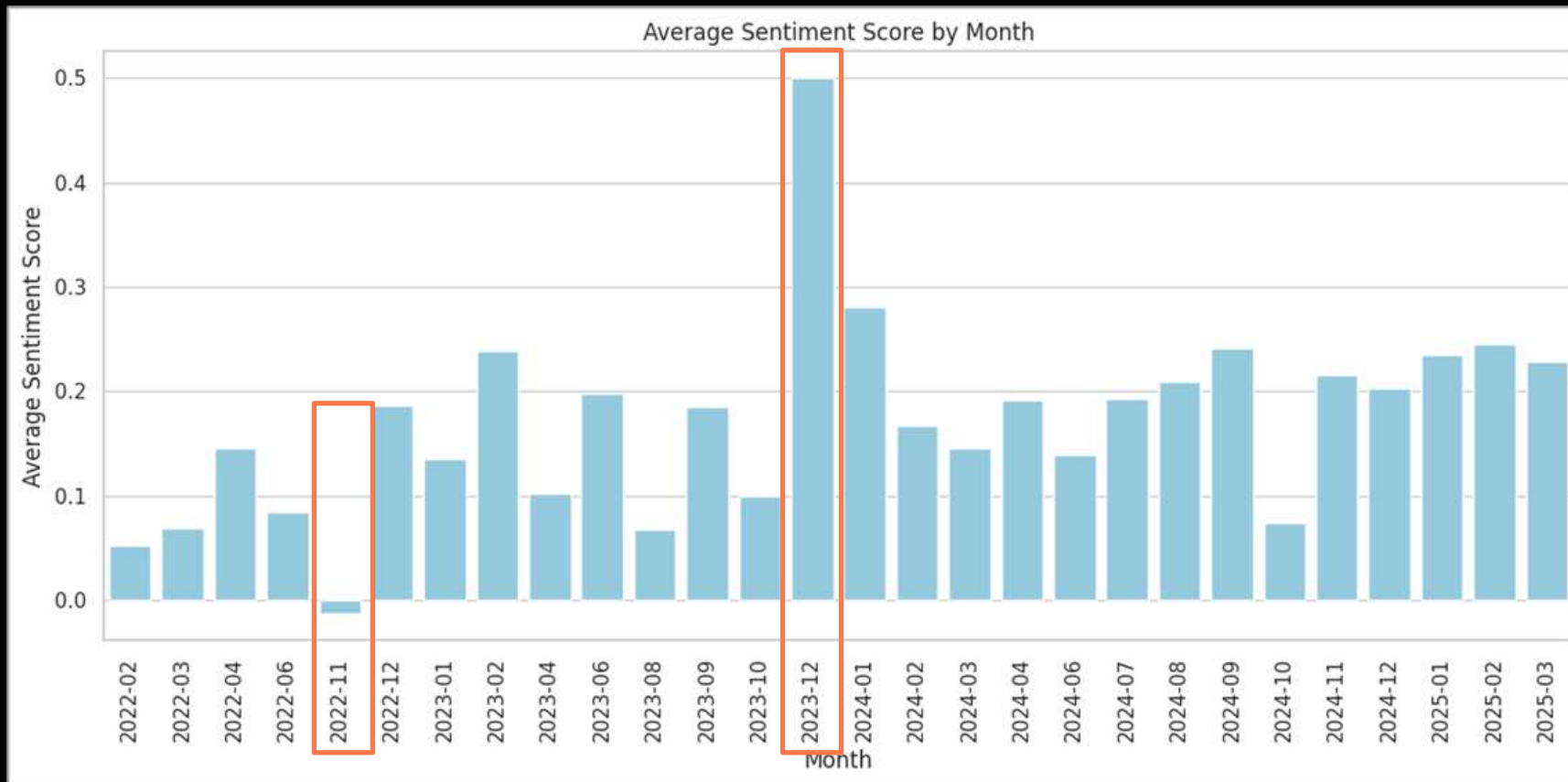
We used two different models pre-trained Machine Learning models to uncover patterns and identify the areas of potential opportunities for improvement.

Model Type	Python Library	Description	Why it was chosen	How it was used	Reference Accuracy*
Sentiment Classifier	NLTK (VADER)	Rule-based sentiment analysis optimized for social media text	Ability to interpret informal language, slang, and emojis, making it ideal for Reddit data	Classified comments as Positive, Neutral, or Negative to identify customer sentiment trends	~0.07* (Precision)
Topic Model (Unsupervised)	BERTopic	Clustering model using BERT embeddings and TF-IDF to identify coherent topics	Ability to capture sentence-level context and group similar discussion themes effectively	Applied to sentiment-filtered comments to uncover the most discussed positive and negative topics related to Nike	0.52–0.68* (Coherence)

**Precision Sources (original whitepapers):
VADER – ²(Hutto & Gilbert, (2014)
BERTopic – ³(Maarten G, (2022)*

Sentiment Classifier

Our Sentiment analysis revealed two important findings.



Sentiment Classifier

Date	Significance	Reason	Interpretation	Our Action
November 2011	Lowest Sentiment Score	#1 Nike's breakup with Kyrie Irving	Customers did not agree with this change. Showing that some customers put a lot of weight on Nike's sponsorships	Review current sponsorships and consider new well-liked athletes
		# 2 Sewing error on expensive (\$100) hooded sweatshirt	Quality control is especially important for higher priced items	Identify case of specific case and review current quality control processes
December 2023	Highest Sentiment Score	AI generated Nike advertisement	Advertisement that gained a lot of positive feedback & engagement	Sales team to further investigate it's impact on sales and reason behind it.

Topic Model (Unsupervised)

The BERTopic Topic Modeling Model uncovered many more actionable insights to the brand.

Sentiment	Topic Cluster	Main Theme	Interpretation	Our Action
Positive	0	Shoe Brands	Customers preferred Nike over Rebook and mentioned compared Nike to Rebook more than any other brand	Shift focus to other top competitors such as Adidas
	1	Kits	Customers prefer Adidas kits over Nikes. Complaining about quality	Look at ways to improving Nike kit quality and get a popular sports team to use them
	2	Sizing	Nike is doing well at producing items that are true to size and fit.	Ensure current process of sizing and fitting continues
Negative	0	Crime	Theft of Nike branded items. Nike is a sought-after brand for re-sell.	None related to improving customer sentiment
	1	MLB	Customers do not like Nike's relationship MLB and its athlete.	Investigate Nike's involvement with MLB and its athletes
	2	Running Shoes	"Most Nikes these days are fashion shoes" and not for runners	Start offering better shoes for running and take less focus off the aesthetics for select running shoes

Challenges & Limitations

Identifying our boundaries

Challenges & Limitations

While we uncovered many items to address, we still need to understand the challenges and limitations of our work.

Challenges

Nike Isolation: Difficult to isolate Nike-specific sentiment due to overlap with competitors.

Language: Informal language, sarcasm, and off-topic content affect sentiment accuracy.

Limitations

Demographics: No demographic data limits segmentation by age, gender, or location.

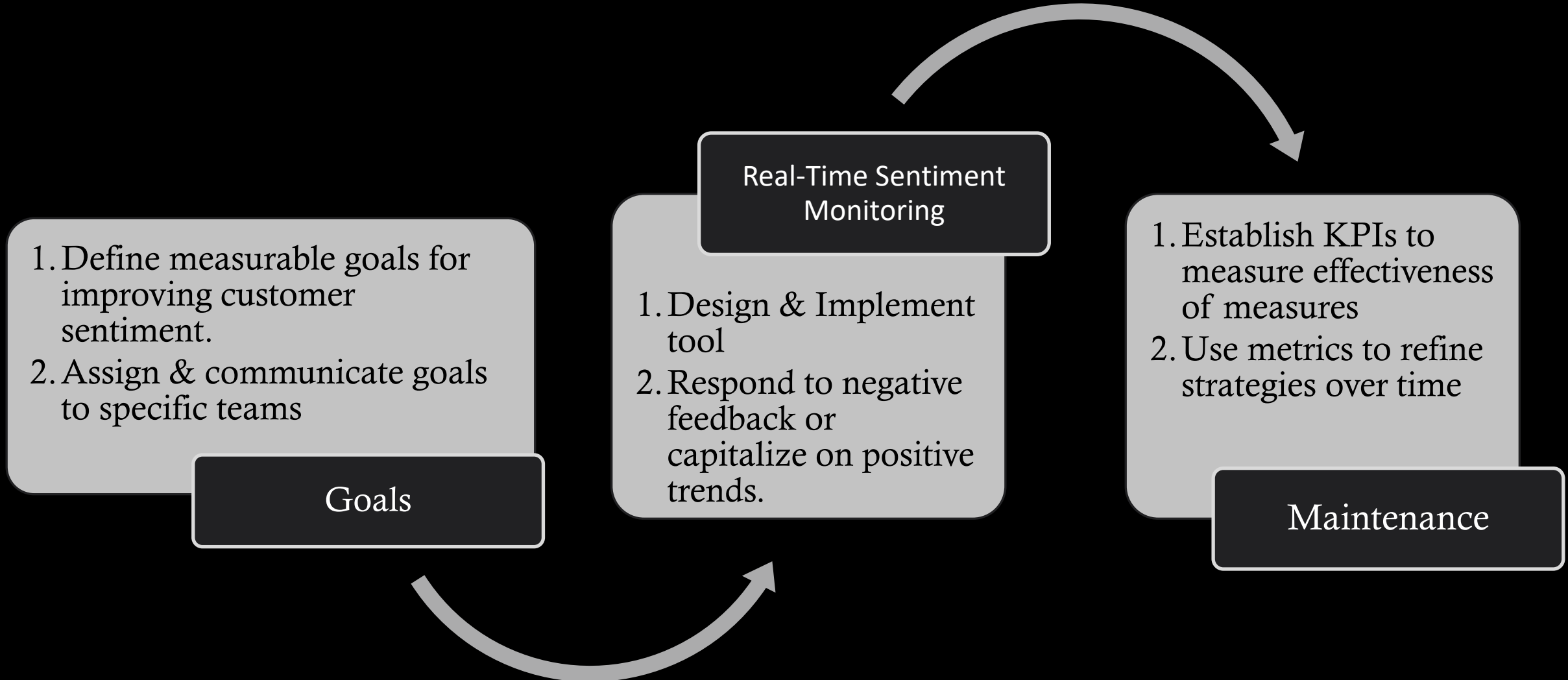
Global Application: Reddit's young, U.S.-centric user base may not reflect global sentiment

External Variables: Sentiment shifts caused by external events are difficult to isolate.

Implementation Plan

Putting our findings to use

Implementation Plan



Ethical Assessment

Our Responsibility

Ethical Assessment

Even though we uncovered many meaningful items and actions to take for improvement, there are several ethical considerations that must be addressed to ensure transparency of our findings.

Ethical Consideration	Key Points
User Privacy	Data must be anonymized with no PII collected. Adhere strictly to Reddit’s API terms and comply with GDPR (EU) and CCPA (California).
Data Bias & Representativeness	Reddit skews young and U.S.-based, which may not reflect Nike’s global audience. Transparency about these demographic limitations is essential to avoid overgeneralized conclusions.
Sentiment Misclassification	Tools like VADER may misinterpret sarcasm, slang, or nuanced emotion, leading to inaccurate sentiment labeling. Nike should avoid over-reliance on automated classifications and adjust strategies accordingly.
Transparency	Nike must be open about the data collection, processing, and analysis methods. Accountability for decisions based on these insights is critical, especially when they impact customer-facing functions.
Impact on Vulnerable Groups	Targeted marketing based on sentiment insights should not exploit vulnerable consumers. Ethical marketing practices and open feedback channels should be implemented to mitigate harm.

Q&A

Opening the floor

Q&A

- Floor is open for any questions for the remainder of the call.
- A list of pre-determined questions will be sent via email to all participants.
- Follow up questions can also be sent to the presenter's email.



Sources

1. NIKE, Inc. (NKE) Revenue by segment. (n.d.). StockAnalysis.
https://stockanalysis.com/stocks/nke/metrics/revenue-by-segment/?utm_source=chatgpt.com
1. Hutto, C., & Gilbert, E. (2014). VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216-225. <https://doi.org/10.1609/icwsm.v8i1.14550>
2. Maarten G, (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. Ar5iv. <https://doi.org/10.48550/arXiv.2203.05794>